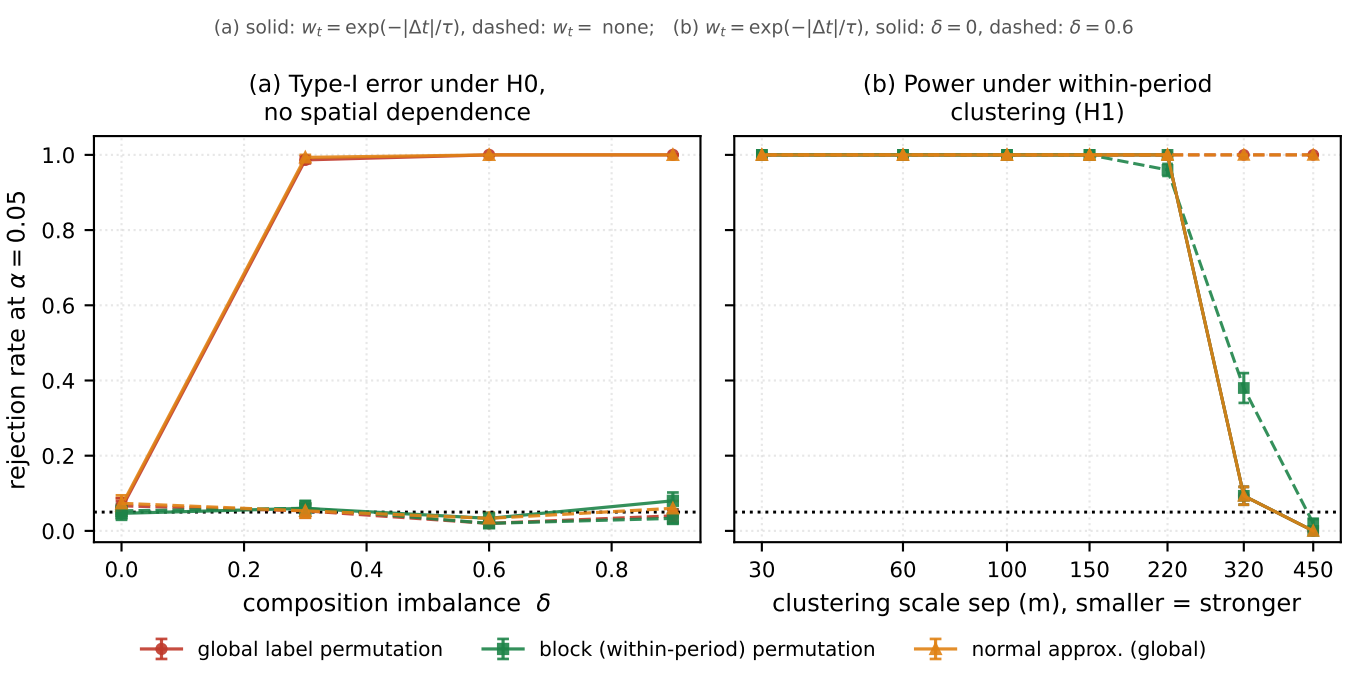
(a) solid: $w_t = \exp(-|\Delta t|/\tau)$, dashed: $w_t =$ none; (b) $w_t = \exp(-|\Delta t|/\tau)$, solid: $\delta = 0$, dashed: $\delta = 0.6$
(a) Type-I error under H0, no spatial dependence
(b) Power under within-period clustering (H1)
rejection rate at $\alpha = 0.05$
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
composition imbalance δ
30
60
100
150
220
320
450
clustering scale sep (m), smaller = stronger
global label permutation
block (within-period) permutation
normal approx. (global)